Identification of common extracellular RNA biomarkers across Neurodegenerative and Cognitive disorders

DeCoDe Institute Capstone Event
exRNA team
July 30, 2026

Main Research question

Can common extracellular RNA signatures be identified across Alzheimer’s disease, Parkinson’s disease, mild cognitive disorder, and Lewy body dementia that could serve as shared biomarkers of neurodegeneration?

Key objectives

- **Identify the differentially expressed extracellular RNAs** in AD, PD, PDD, MCD, and LBD.

- Identify **common exRNA signatures** associated with neurodegeneration.

- Assess if the identified exRNAs are associated with key neurological function

AD : Alzemeirs Disease, **PD**: Parkinsons disease, PDD:Parkisons disease with Dementia, **MCD** : Mild cognitive disorder, **LBD**: Lewy body disease

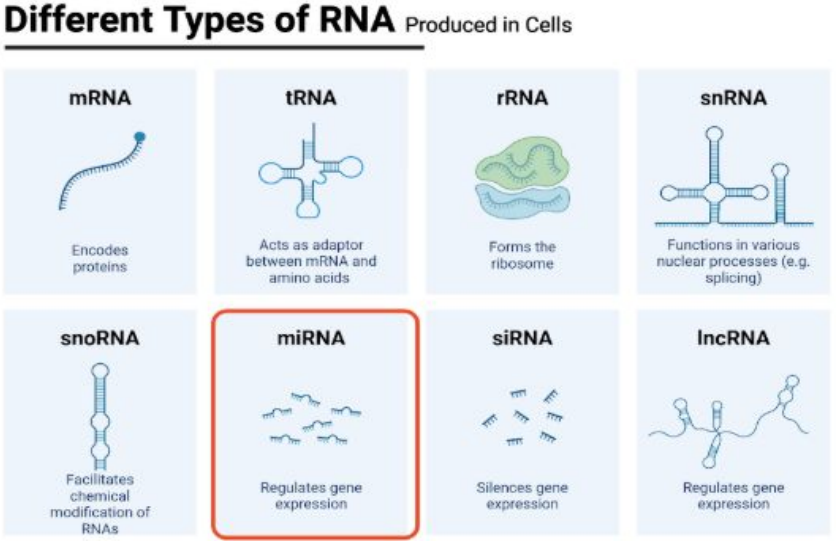
Background

exRNAs are diverse RNA species present outside of cell in various biofluids—such as blood, cerebrospinal fluid (CSF), saliva, and urine.

Plasma source

Stable

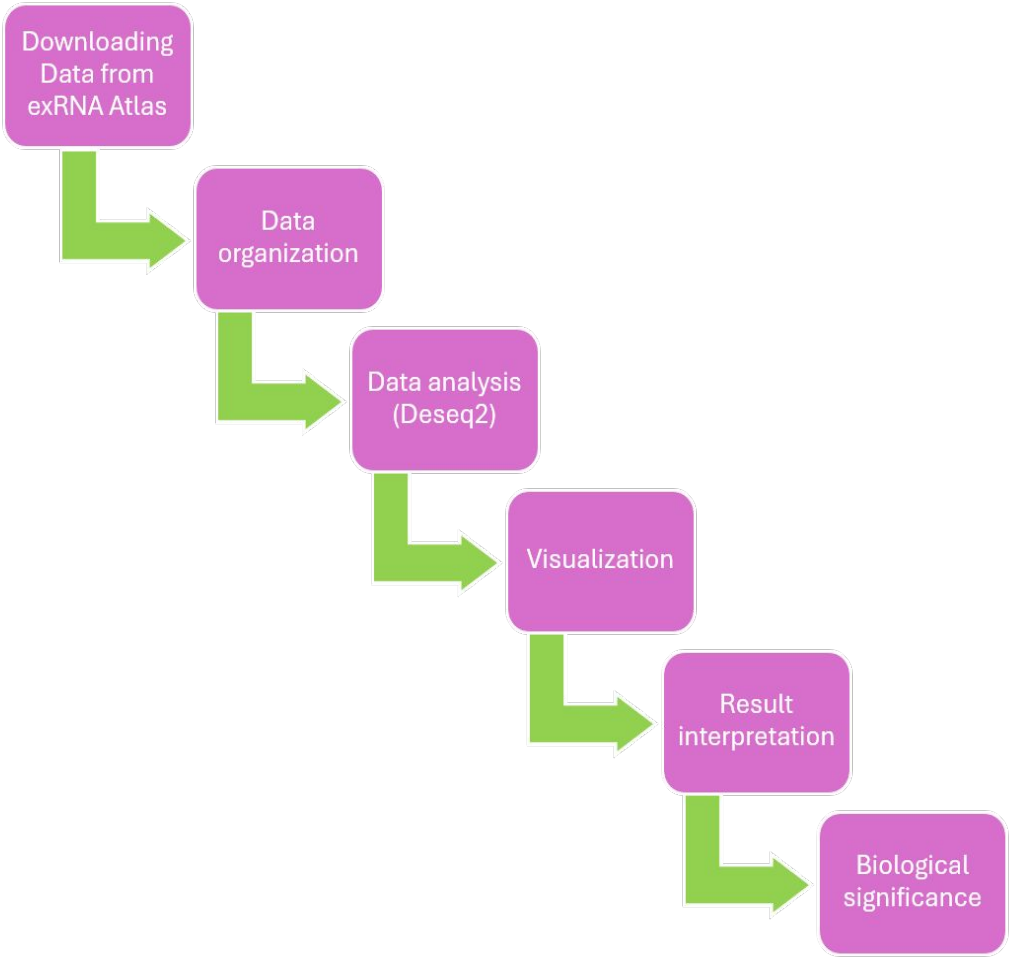
Minimally invasive



Regulate many gene
Upstream of disease
EV enriched, miRNA therapeutics
Sensitive enough to measurable techniques

Methods

Workflow


Downloading Data from exRNA Atlas
Data organization
Data analysis (Deseq2)
Visualization
Result interpretation
Biological significance

Data Source

- **Source:**
  exRNA Atlas, NIH Common Fund
- **Study:**
  KJENS1PDD, Plasma small RNA-seq
- **exceRpt pipeline**
  Uniformly processed (hg38, v5.0.0)
- **Cohort:**
  Neurodegenerative conditions + healthy controls

Condition	n
Healthy Control	54
Alzheimer's Disease (AD)	41
Mild cognitive disorder	41
Parkinson's Disease (PD)	39
Dementia associated with PD	27
Lewy Body Dementia	19

Data Extraction & Preprocessing

Extract counts

Pulled *mature miRNA* from each sample archive;
cleaned IDs;
used raw total ReadCount

→

Remove Duplicates

Some donors appeared multiple times (different kits / pipeline reruns)
Kept one library per donor

→

Merge to one Matrix

Combined all samples into a single miRNA × sample count matrix, joined on miRNA name

Resulting Matrix: **1,976** miRNAs × **221** samples

Differential Expression Analysis

Input

Raw Counts (miRNA_matrix)

Filter & QC

Remove low-count miRNAs (≥ 10 total read)

Library sizes, normalized-count distributions, PCA per comparison

DESeq2

Shrinkage applied for the effect sizes
Stabilized with ashr

Comparisons:

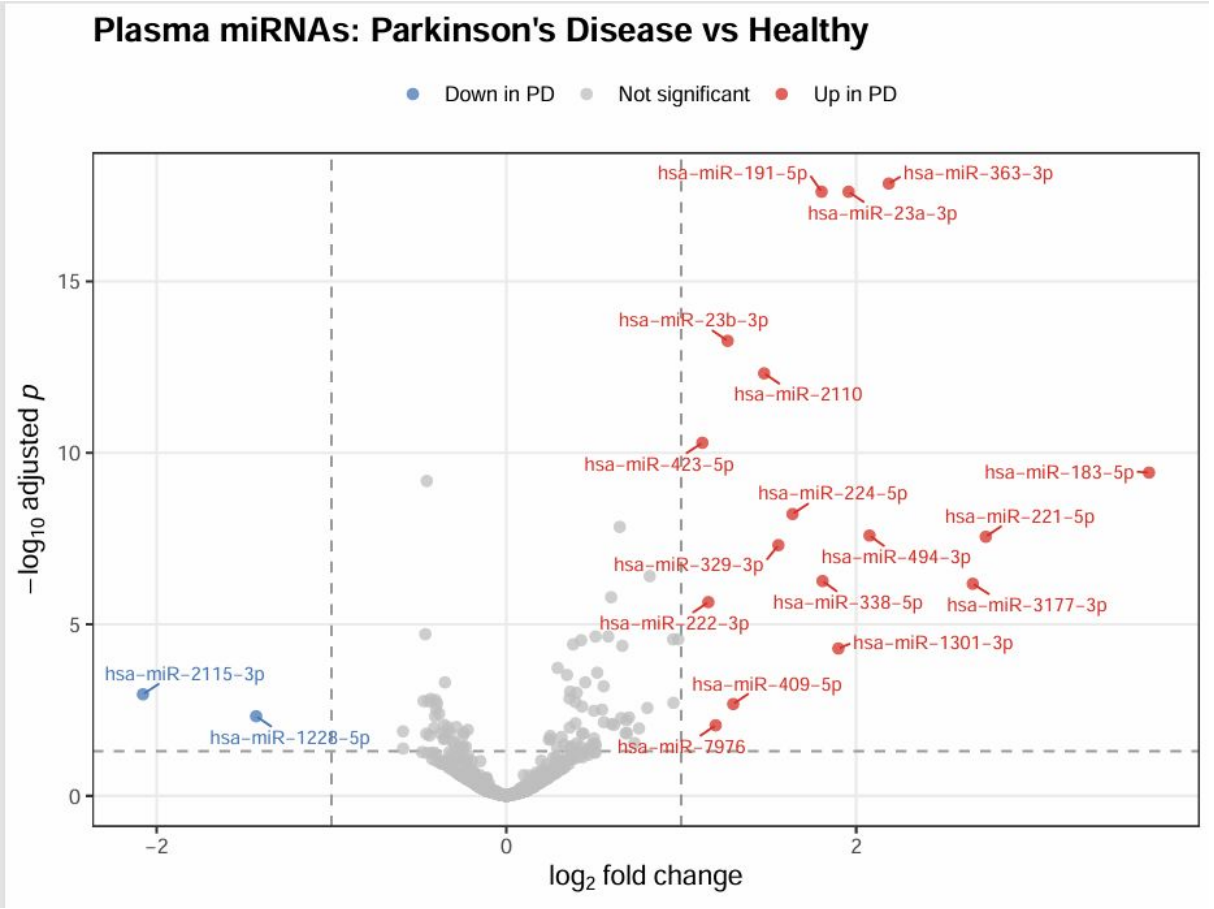
1. **Each condition vs. Healthy**
2. **All disease pooled vs. Healthy**

Biomarker Threshold: **P < 0.05 & |log2FC| > 1**

RESULTS

Results

- 5 individual DE analysis for each disease vs healthy controls
- 1 analysis combining samples from all diseases vs health controls



AD : Alzemeirs Disease, **PD**: Parkinsons disease, PDD:Parkisons disease with Dementia, **MCD** : Mild cognitive disorder, **LBD**: Lewy body disease

Identified 22 DE miRNA across all analysis

Upregulated miRNAs

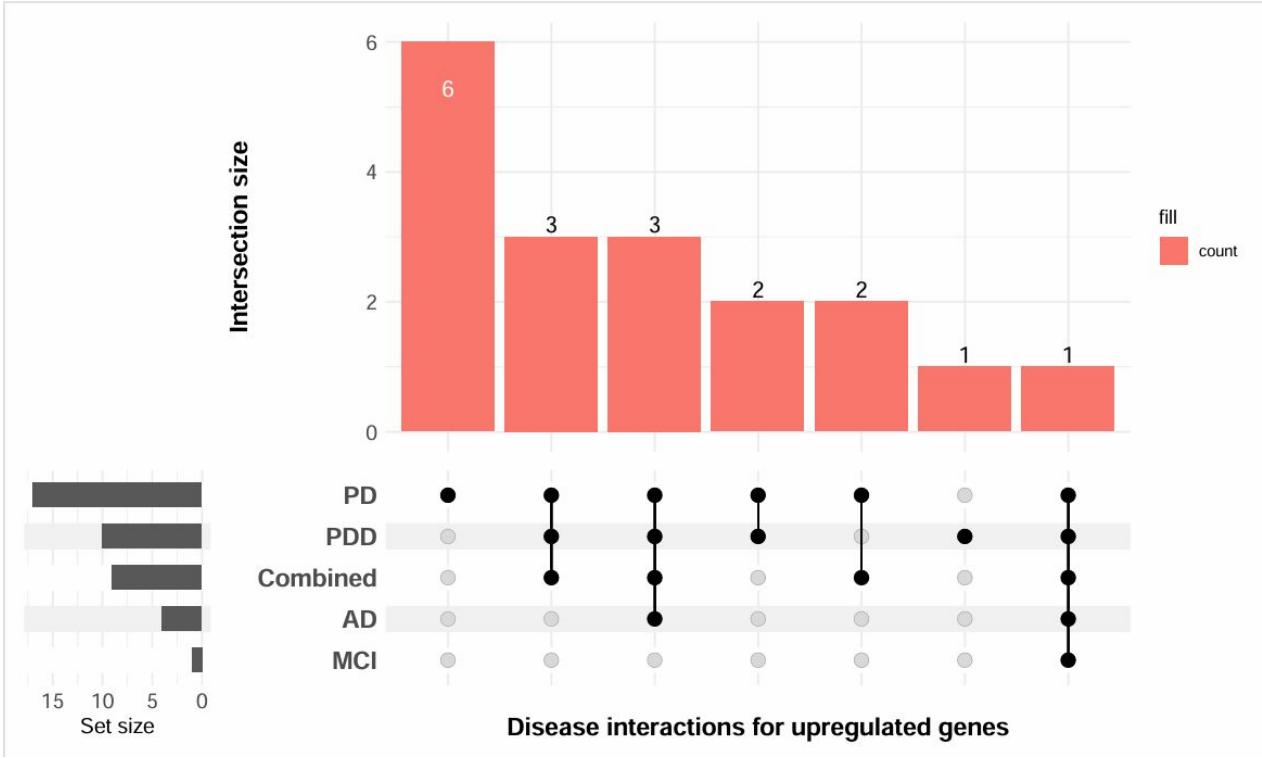
- 18 unique miRNAs are differentially upregulated
- Majority of the miRNAs are shared across one or more diseases
- More on next slide ….

Downregulated miRNAs

- 4 unique downregulated miRNAs identified in PD(2), PDD(1), LBD(1)
- There is no overlap across down regulated miRNAs

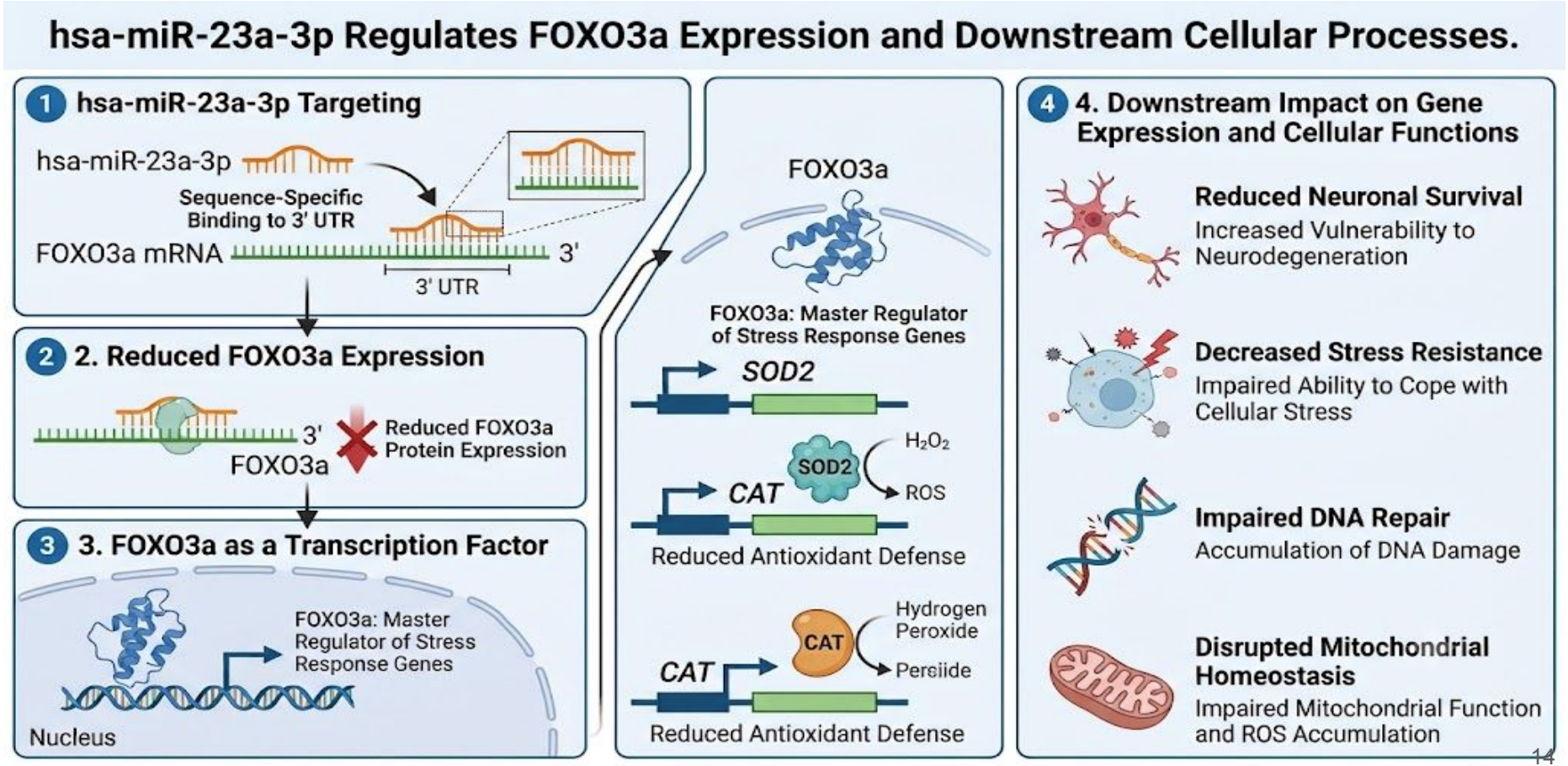
hsa-miR-23a-3p is a common biomarker across diseases (except LWD)

- hsa-miR-23a-3p is DE across all diseases except LWD
- PD and PDD have unique miRNA candidates



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Prime candidate miRNA



Wang, S., Sun, Y., Yao, L. *et al.* The Role of microRNA-23a-3p in the Progression of Human Aging Process by Targeting FOXO3a. *Mol Biotechnol* **66**, 277–287 (2024). https://doi.org/10.1007/s12033-023-00746-7

Conclusion

- **Identify the differentially expressed miRNAs** in AD, PD, PDD, MCD, and LBD.
  - We identified 22 DE miRNAs

- Identify **common exRNA signatures** associated with neurodegeneration.
  - We identified one common miRNA across all diseases (except LBD)

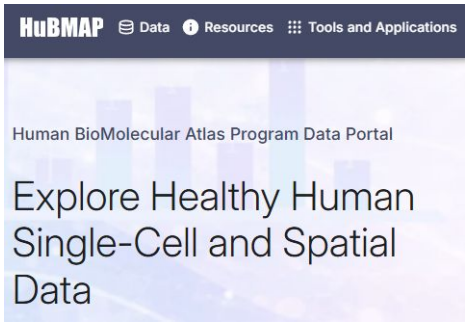
All the plots, code and data are deposited on GitHub

AD : Alzemeirs Disease, **PD**: Parkinsons disease, PDD:Parkisons disease with Dementia, **MCD** : Mild cognitive disorder, **LBD**: Lewy body disease

Challenges

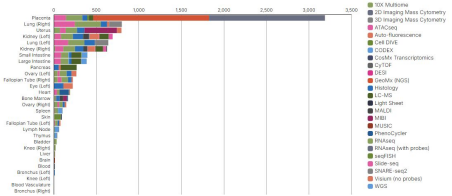
Capstone

Day 1	Day 2	Day 3





- *All healthy*

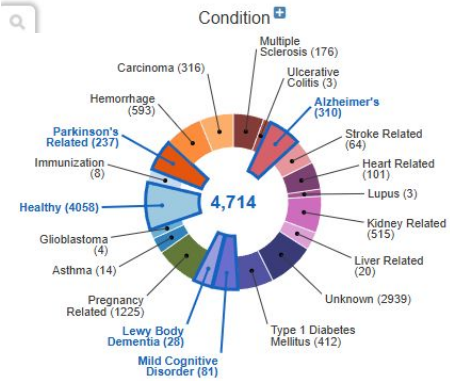


- *No access*
- *Very large files for spatial transcriptomics*



-No batch download
-Low samples
-Diff. Sample Processing
-MS samples (initial choice) were qPCR vs other were small RNA-seq

exRNA



-Genome Migration: The Atlas was mid-migration so older download files were unavailable during the transition between reference genomes.

-Network Restrictions: Standard sample download files were blocked by firewalls.

-Access Barriers: Built-in analysis tools required consortium-level access and were unavailable.

What We Will Take Forward from the DeCoDE Workshop

- Complexity and diversity of NIH common fund data and how to navigate it.
- Rekindle confidence in coding and the joy of it!
- Open access does not necessarily mean easy access.
- Metadata is key to identifying the working set of samples
- Your first question may not be the final question you are answering.
- Science is a collaborative effort; compiling diverse research builds solid biological conclusions.

Resources

Resources:

1. Wang, S., Sun, Y., Yao, L. *et al.* The Role of microRNA-23a-3p in the Progression of Human Aging Process by Targeting FOXO3a. *Mol Biotechnol* **66**, 277–287 (2024). https://doi.org/10.1007/s12033-023-00746-7
2. https://exrna.org/
3. https://portal.hubmapconsortium.org/
4. https://gtexportal.org/home/

AI use: AI applications were primarily used for literature review, help with code syntax

- nano banana 2
- Gemini

Biorender

- Illustrations

THANK YOU & QUESTIONS

- Deepti Jain
- Marina Rice
- Marangelie Criado-Marrero
- Munira Haque
- Nur Shahir
- Zaynab Shakkour

Special thanks to CFDE training center for this training opportunity and guidance

CFDE Training Center
Decoding the Data Ecosystem

DeCoDE
INSTITUTE
SUMMER 2026



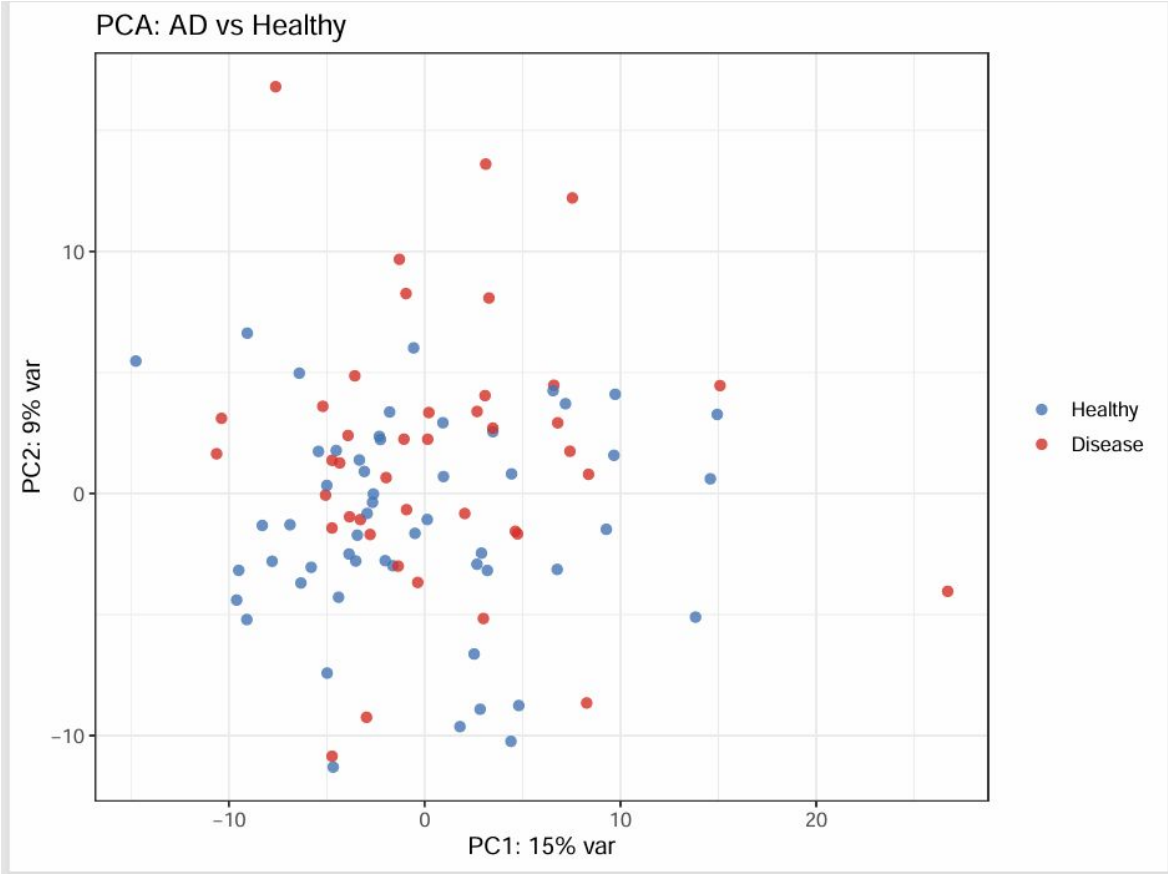
Kelli Bursey



Allissa Dillman

EXTRA slides

QC


PCA: AD vs Healthy
PC2: 9% var
PC1: 15% var
10
0
−10
−10
0
10
20
Healthy
Disease